# Public norm jets and coupled branch reconstruction

2026-11-13. Unsigned producer evidence [pcc:unverified].

**Question.** Can explicit node-preserving polynomial jets correct a norm while retaining the pole-degree window? This round tests a specified polynomial family, without claiming it represents the older matrix $W$-family.

**Exact map.** Put $N(A,B) = A^2 - fB^2$, $A' = A + GJ$, $B' = B + GK$. Then

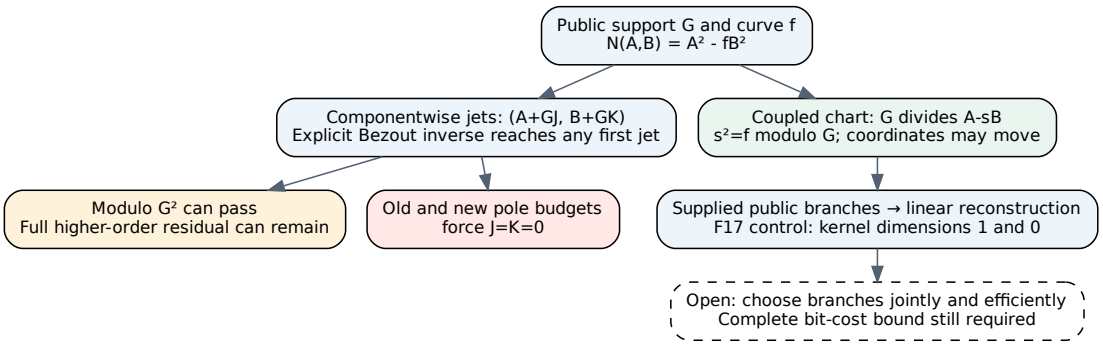
$$N(A',B') = N(A,B) + G(2AJ - 2fBK) + G^2(J^2 - fK^2).$$

For $2u = 1$ and $AU + fBV + GW = 1$, take $J = uU\Delta$, $K = -uV\Delta$. Exactly

$$N(A',B') - N(A,B) - G\Delta = G^2(J^2 - fK^2 - W\Delta).$$

Every first quotient jet is reachable with the explicit inverse, but the full residual is retained. This is classical Hensel algebra [pcc:unverified].

**Actual degree constraint.** If old and new coordinate degrees are all strictly below $\deg G$, differences divisible by $G$ are zero. Thus $J = K = 0$ when $G \neq 0$. The actual pole bounds $2\deg A \leq \deg G$ and $\deg f + 2\deg B \leq \deg G$ imply those strict bounds when $\deg G > 0$. Componentwise jets cannot produce a new target norm inside that window. This conclusion does not exclude coupled or rational operations [pcc:unverified].




**Coupled successor.** Fix $s$ with $G \mid s^2 - f$. Changes $(A + sH + GJ, B + H)$ preserve $A - sB$ modulo $G$ while individual coordinates can move. Since

$$A^2 - fB^2 = (A - sB)(A + sB) + (s^2 - f)B^2,$$

branch congruence gives norm divisibility. Monic $G$, actual pole bounds and nonzero norm then give $N = CcG$ for $c \neq 0$, without moments or cofactors. Choosing the root branches and finding the bounded pair remain inputs [pcc:unverified].

## Fixed controls, proof scope and next finder obligation

**Local versus exact.** One write-once native fixture tests all289 $\Delta$ of degree at most1 over F17, with $f = X^3 + 2X + 2$, $G = X$, $A = 6$, $B = 1$, $V = 9$, $W = 16 + 8X^2$. Exactly288 retain a nonzero full residual. For target $c = 1$, the public lift gives

$$B' = 1 + 12X + 4X^3, \qquad N(6, B') - G = 4X^2 + X^3 + 6X^4 + 4X^5 + 11X^6 + 8X^7 + X^9.$$

It passes modulo $G^2$, but fails the exact norm and degree window. This is an algebra fixture of support degree1, not a cubic ECDLP relation instance. Coefficient identities, node/derivative values and all17 point evaluations are checked; cases are deterministic, not independent trials [pcc:unverified].

**Public branch reconstruction.** From the supplied public points $(0, 6), (3, 1), (5, 1), (6, 3), (10, 11), (13, 7)$, construct $s$ by Lagrange interpolation and solve the six original homogeneous equations for $\deg A \leq 3$, $\deg B \leq 1$. No known $A/B$ is supplied to that solve. Check original rows, point equations, branch congruence, degree bounds and full norm identity [pcc:unverified].

| Supplied assignment | Kernel dimension | Result |
|---|---|---|
| Six points above | 1 | $A = [14, 2, 7, 3],\ B = [8, 1],\ c = 9$ |
| Only $(13, 7) \to (13, 10)$ | 0 | No bounded nonzero pair for these branches |

The positive pair is3 times the original fixture pair, with norm scaling9. The flipped result concerns one sign assignment; it does not exclude the unsigned support, whose positive assignment is exhibited. This is supplied-branch reconstruction, not branch selection or an attack search [pcc:unverified].

**Formalization.** Fourteen new statements in `NormPublicJets.lean`; source `8892e4477`, final native source `a5b04b3e5`. Whole-study tracked-source gate:231 statements,0 additional trusted axioms. Explicit kernel replay receipts and source hashes accompany the report. Lean4.35.0, CryptoLean`148f5dc389b`, Mathlib`cfc55fc58714`. Initial build001 failed on a lemma invocation;002 passes. Strict clippy001 failed on an indexed loop;002 passes after an actual repair, with7 tests passing. All failures and exact exits remain [pcc:unverified].

**Source claims, unsigned.**

First jet: `PCC-20261113-4969cb`.

Actual-window implication: `PCC-20261113-f8bb46`.

Coupled certificate: `PCC-20261113-41d872`.

Native fixture: `PCC-20261113-e63c86`.

The full14-statement proof map and claim pins are in the source-linked README. Default-PATH source-only PCC checks are not_covered, distinct from the explicit-toolchain kernel checks. No signed verdict is supplied.

**Next obligation.** Efficiently select square-root branches jointly with bounded reconstruction, with all rejected attempts and setup charged. Keep $\exp(o(\log r))$ complete bit cost on general prime curves as the acceptance criterion. The finite linear solve does not discharge branch selection [pcc:unverified].

**Prior art.** Galbraith, *Mathematics of Public Key Cryptography*, 2.13 (Hensel expansion),8.4 (Riemann–Roch/pole constraints), cached primary text; previous norm derivation and recovery source. KB MCP unavailable; canonical wrapper exits127 (no executable). No exhaustive or novelty claim.